

Unit 1:Introduction

- 1.1. Definition
- 1.2. History of IoT
- 1.3. IoT Architecture
- 1.4. IoT Frameworks
- 1.5. Benefits of IoT
- 1.6. Applications of IoT

IoT

IoT describes the network of physical objects called things that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.

IoT refers to a system of interrelated, internet-connected objects that are able to collect and transfer data over a wireless channel without human intervention.

IoT enables to analyze and act on data, facilitating the decision making process with aid of insights obtained.

Examples:

- Smart watch
- Smart speaker
- Smart camera
- Smart lights
- Smart thermostats
- Wearable fitness trackers
- Industrial sensors and smart appliances
- Smart sensors for environmental monitoring
- Connected home devices smart locks, thermostats
- Smart TV remotes
- Smart home remotes
- Remote car keys
- Industrial IoT remotes

Characteristics of IoT:

The Internet of Things (IoT) has several key characteristics that define its functionality and impact. Some of them are as follows

Connectivity: IoT devices are interconnected via networks like Wi-Fi, Bluetooth, or cellular, enabling seamless communication and data exchange.

Sensing: IoT devices use sensors to collect data from the physical environment, such as temperature, motion, or humidity.

Data Processing: The collected data is processed either locally (on the device) or in the cloud to derive meaningful insights or trigger actions.

Automation: IoT enables devices to perform tasks automatically without requiring human intervention, based on predefined rules or real-time data.

Scalability: IoT systems can grow and support many devices, accommodating a vast network of interconnected devices.

Real-Time Operation: IoT facilitates real-time monitoring, communication, and decision-making for time-sensitive applications.

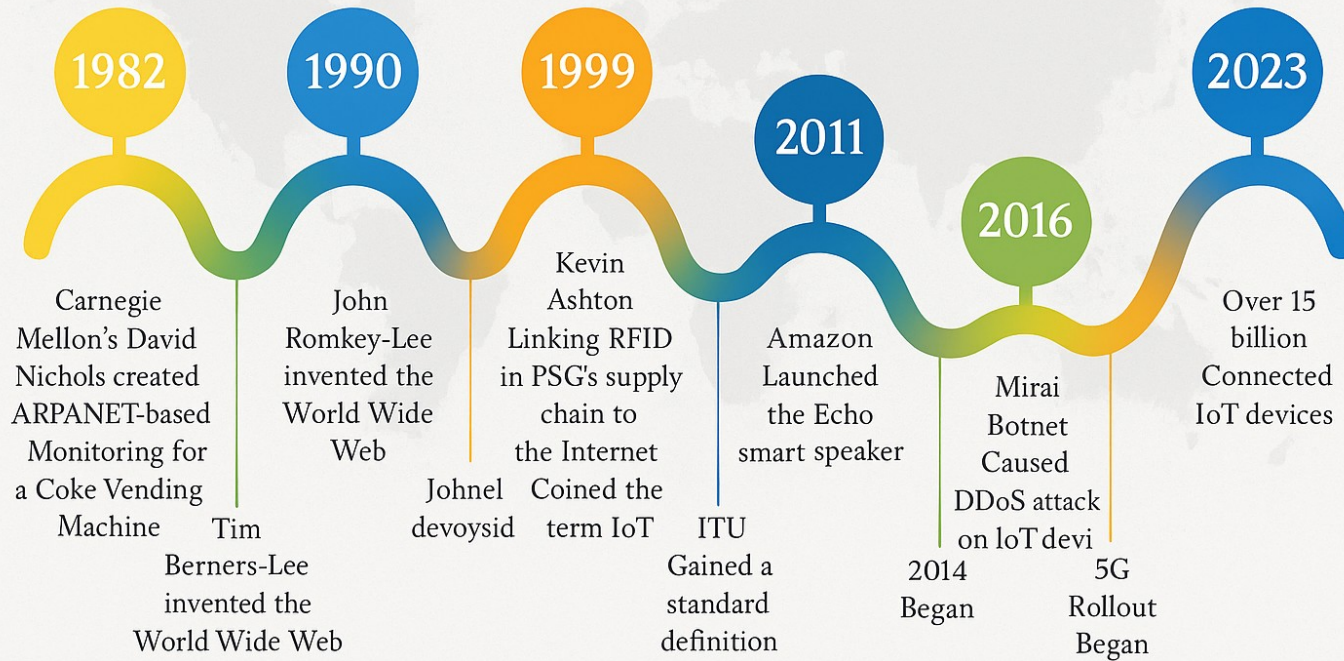
Remote Control and Access: Devices can be managed and controlled remotely, offering convenience and efficiency.

Interoperability: IoT systems are designed to work across different platforms, devices, and technologies, ensuring smooth collaboration.

Data-Driven Decisions: By analyzing data, IoT enhances decision-making processes in industries like healthcare, agriculture, and transportation.

Security and Privacy: As IoT devices handle sensitive data, robust measures are implemented to ensure cyber security and protect user privacy.

History of IoT



History of IoT

1982 – Carnegie Mellon Vending Machine

It is considered the first internet-connected appliance, designed to monitor drink availability and temperature remotely via ARPANET.

1989 – Tim Berners-Lee and the World Wide Web

Tim Berners-Lee did invent the World Wide Web in 1989, which became the key platform over which IoT devices could communicate.

1990 – John Romkey's Toaster

John Romkey built the first Internet-connected toaster that could be switched on and off remotely using TCP/IP.

1999 – Kevin Ashton Coined "IoT"

Kevin Ashton proposed the idea of connecting RFID to the internet and coined the term "Internet of Things" while working at Procter & Gamble.

2008 – IoT Recognized Internationally

A significant IoT conference was indeed held in Zurich.

The US National Intelligence Council identified IoT as a Disruptive Civil Technology, officially recognizing its transformative potential.

2011 – IoT Gains Standard Definition

The International Telecommunication Union (ITU) published a formal definition of IoT. This provided global clarity on what IoT is and helped with standardization efforts.

2016 – Mirai Botnet Attack

- The Mirai malware infected thousands of poorly secured IoT devices and launched a massive DDoS attack on DNS provider Dyn. This incident Highlighted serious security vulnerabilities in IoT systems.

2019 – 5G Rollout Begins

The global rollout of 5G networks began.

5G dramatically enhances IoT connectivity, enabling faster data transfer and ultra-low latency.

2020–2021 – IoT in Healthcare During COVID-19

Remote patient monitoring, smart thermometers, and contact tracing apps became

Widespread. It showed the real-world value of IoT in public health and crisis response.

2023 – Over 15 Billion Connected Devices

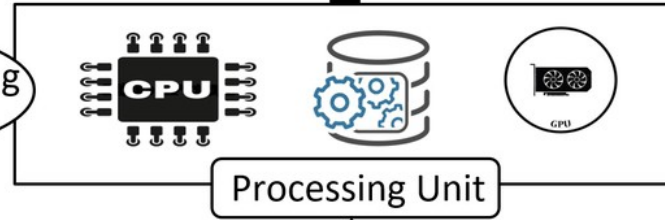
IoT Analytics reported that IoT devices worldwide surpassed 15 billion.

This demonstrates exponential growth and integration into daily life and industries.

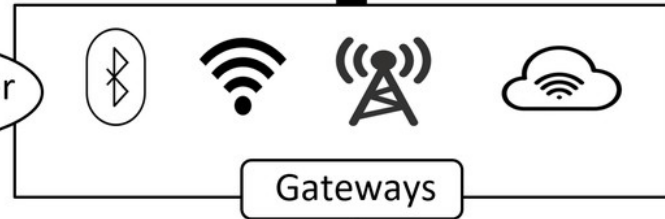
Application
Layer



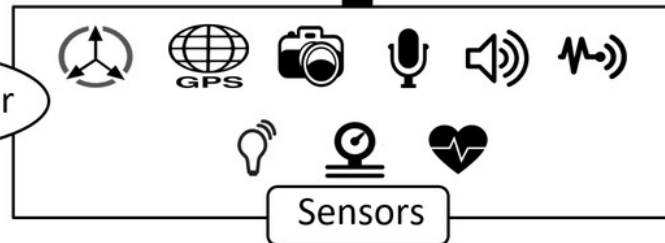
Data Processing
Layer



Network Layer



Sensing Layer



IoT Architecture:

Sensing Layer:

The sensing layer is the first layer of the Internet of things architecture and is responsible for collecting data from different sources. This layer includes sensors and actuators that are placed in the environment to gather information about temperature, humidity, light, sound, and other physical parameters. Wired or wireless communication protocols connect these devices to the network layer.

Network Layer: The network layer of an IoT architecture is responsible for providing communication and connectivity between devices in the IoT system. It includes protocols and technologies that enable devices to connect and communicate with each other and with the wider internet. Examples of network technologies that are commonly used in IoT include WiFi, Bluetooth, Zigbee, and cellular networks such as 4G and 5G technology. Additionally, the network layer may include gateways and routers that act as intermediaries between devices and the wider internet, and may also include security features such as encryption and authentication to protect against unauthorized access.

Data processing Layer: The data processing layer of IoT architecture refers to the software and hardware components that are responsible for collecting, analyzing, and interpreting data from IoT devices. This layer is responsible for receiving raw data from the devices, processing it, and making it available for further analysis or action. The data processing layer includes a variety of technologies and tools, such as data management systems, analytics platforms, and machine learning algorithms. These tools are used to extract meaningful insights from the data and make decisions based on that data. Example of a technology used in the data processing layer is a data lake, which is a centralized repository for storing raw data from IoT devices.

Application Layer: The application layer of IoT architecture is the topmost layer that interacts directly with the end-user. It is responsible for providing user-friendly interfaces and functionalities that enable users to access and control IoT devices. This layer includes various software and applications such as mobile apps, web portals, and other user interfaces that are designed to interact with the underlying IoT infrastructure. It also includes middleware services that allow different IoT devices and systems to communicate and share data seamlessly. The application layer also includes analytics and processing capabilities that allow data to be analyzed and transformed into meaningful insight.

Advantages of IoT:

1. Efficiency and Automation

IoT automates routine tasks, reducing manual intervention and increasing efficiency. For example, smart thermostats adjust temperatures based on occupancy, saving energy and reducing costs.

2. Enhanced Decision-Making

Real-time data and advanced analytics enable informed decision-making. In industrial settings, IoT systems can predict equipment failures, allowing for proactive maintenance and minimizing downtime.

3. Improved Quality of Life

IoT enhances daily life by providing convenience, security, and health benefits. Smart home devices simplify tasks, while wearable health monitors provide critical health insights.

4. Cost Savings

By optimizing processes and reducing waste, IoT can lead to significant cost savings. Smart grids, for instance, balance electricity supply and demand, reducing energy costs.

Disadvantages:

Security: The data is traveling all over the Internet. So maintaining its privacy is still a Big Challenge. End-to-end Encryption is a must in IoT.

Compatibility: There is no International Standard for the monitoring of the equipment.

Complexity: Most of the devices still contain some software bugs.
Each device must be able to seamlessly interact with other devices in the network.

Safety: If an IoT device fails or gives incorrect data , it can lead to unsafe conditions without time

Applications of IoT:

1. Smart Homes

IoT transforms homes into smart living spaces, enhancing convenience and security.

Examples include:

Smart Lighting: Automated and remote-controlled lighting systems.

Security Systems: Surveillance cameras, smart locks.

Energy Management: Smart thermostats, energy monitoring devices.

2. Industrial IoT (IIoT)

IIoT enhances industrial processes through automation, predictive maintenance, and improved efficiency. It includes

Smart Manufacturing: Connected machinery, real-time monitoring.

Supply Chain Management: Tracking and optimizing logistics.

Predictive Maintenance: Anticipating equipment failures before they occur.

Energy Management: Smart grids, street lighting.

3. Healthcare

IoT in healthcare, or the Internet of Medical Things (IoMT), improves patient care and operational efficiency. Examples include:

Wearable Health Monitors: Tracking vital signs, activity levels.

Remote Patient Monitoring: Keeping track of patients outside hospital settings.

Smart Medical Devices: Connected inhalers, insulin pens.

4. Smart Cities

IoT enables the development of smart cities, enhancing urban living through efficient resource management and improved services. Applications include:

Smart Traffic Management: Reducing congestion with real-time data.

Waste Management: Optimizing c

5. Agriculture

IoT revolutionizes agriculture by increasing productivity and sustainability. Examples include:

Precision Farming: Monitoring soil conditions, crop health.

Livestock Monitoring: Tracking animal health and movements.

Smart Irrigation Systems: Efficient water usage based on real-time data.

IoT Frameworks:

An IoT framework provides a structured approach for designing, implementing and managing IoT system. It typically include component like

Devices and sensor:

These are physical object or sensors which collect data to perform action.

Connectivity:

This define protocols and technologies (eg. for wifi, bluetooth, Zigbee) for communication between devices and other system.

Data Processing:

This performs mechanism for handling, storing and analyzing data, often involving, edge computing and cloud computing

Data Storage:

Database or storage system to keep collected data for future use

Analytics and Intelligence:

It includes tools and algorithm for analyzing data to gain insights or make decision

User Interface:

It includes dashboard or application that allow users to interact and monitor the IoT system

Security:

It involves various measures to protect data and devices from unauthorized access or attacks

Integration:

IT describes the APIs and middleware for integrating IoT system with other enterprise system or application